

Problem

Design an analog computer to simulate

$$\frac{d^2 v_o}{dt^2} + 2 \frac{dv_o}{dt} + v_o = 10 \sin 2t$$

where $v_o(0) = -6$ V and $v_o'(0) = 0$.

Step-by-step solution

Step 1 of 1

$$\frac{d^2 V_o}{dt^2} = 10 \sin 2t - \frac{2dV_o}{dt} - V_o$$

Thus by combining integrators with a summer, we obtain the appropriate analog computer as shown below.

